

Project: Designing 3 Tier Architecture In AWS

Designing a new web application in three tiers. Outline for the project is below:

Web Tier

2 public subnets

Minimum 2 EC2 instances with OS of choice (free tier) in an auto scaling group

EC2 web server security group allowing inbound permission from the internet

Bootstrap static web page or create a custom AMI that already includes the static web page

Create a public route table and associate the 2 public web subnets

Application Tier

2 private subnets

Minimum of 2 EC2 instances with OS of choice (free tier) in an auto scaling group

EC2 application server security group allowing permission from the web server security group

Create a private route table and associate the 2 private App subnets

Database Tier

2 private subnets

Use a free tier MySQL RDS database

The database security group should allow inbound traffic for MySQL from the app server security group

Create a private route table and associate the 2 private database subnets

Before we start:

Let's set up a **Virtual Private Cloud (VPC)**.

Navigate to VPC on the AWS Console and click on "Create VPC".

VPC Settings: VPC only

Name tag: multi-tier-vpc

IPv4 CIDR block: IPv4 CIDR manual input

IPv4 CIDR: 10.10.0.0/16 (Yes I've used this CIDR block before and I'm using it again for familiarity.)

Then click on "create VPC" at the bottom.

[Alt+S]

United States (N. Virginia)
srinadhgdlr

VPC > Your VPCs > Create VPC

Create VPC Info

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create Info
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only
☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

IPv4 CIDR block Info

☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

CIDR block size must be between /16 and /28.

IPv6 CIDR block Info

☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tenancy Info

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

You can add 49 more tags

The VPC is now available and ready.

[Alt+S]

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VPC dashboard <

EC2 Global View
Filter by VPC

Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs

Your VPCs (1/2) Info

Last updated 1 minute ago

Name	VPC ID	State	Block Public...	IPv4 CIDR	IPv6 CIDR
-	yvc-0f982f9178eb91a4f	Available	Off	172.31.0.0/16	-
☒ multi-tier-vpc	yvc-031c57b95dfed4773	Available	Off	10.10.0.0/16	-

vpc-031c57b95dfed4773 / multi-tier-vpc

Details
Resource map
CIDRs
Flow logs
Tags
Integrations

Details

VPC ID
vpc-031c57b95dfed4773

DNS resolution
Enabled

Main network ACL
-

IPv6 CIDR (Network border group)
-

State
Available

Tenancy
default

Default VPC
No

Network Address Usage metrics
Disabled

Block Public Access
Off

DHCP option set
dopt-0ab5c96a33ae0222f

IPv4 CIDR
10.10.0.0/16

Route 53 Resolver DNS Firewall rule groups
-

DNS hostnames
Enabled

Main route table
-

IPv6 pool
-

Owner ID
284514300790

Now to set up our Subnets.

I am setting up a total of 6- 2 public subnets for the web tier, 2 private subnets for the application tier, and 2 private subnets for the database tier.

Navigate to Subnets and click on “Create subnet”.

Select the VPC you just made for the VPC ID, name your subnets in order of what they are for. I went with this web/application/database-tier-public/private-# scheme but of course you can name them however you want.

This project is to have 2 availability zones so let’s select AZ 1a for this first subnet and 1b for the 2nd one. I’m following the order of the outline above so I’m starting with the web tier public subnets first. For the IPv4 CIDR block I went with another familiar scheme- 10.10.1/2/3/4/5/6.0/24.

10.10.1.0/24

10.10.2.0/24

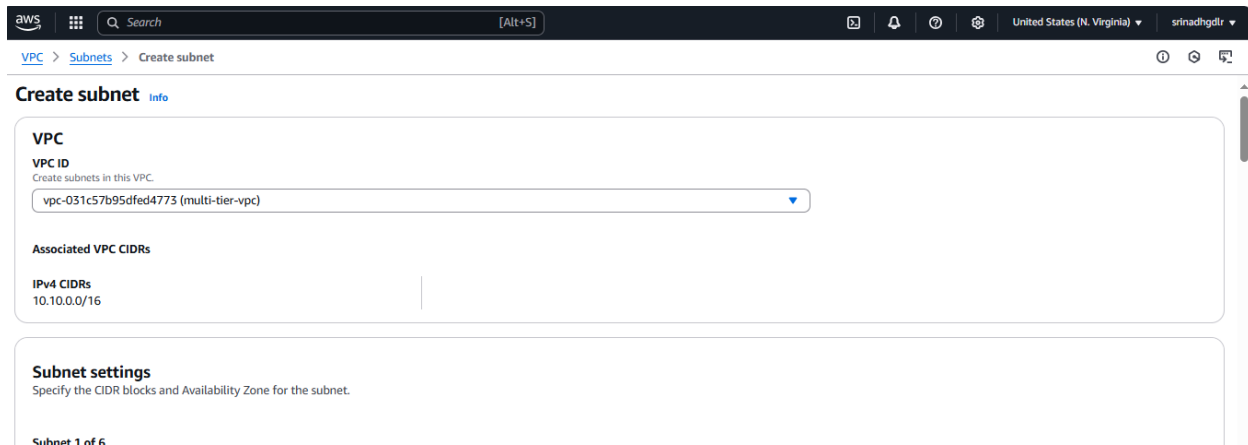
10.10.3.0/24

10.10.4.0/24

10.10.5.0/24

10.10.6.0/24

Instead of clicking “create subnet” at the bottom, I am going to click on “add new subnet” to make the rest of the subnets. See screenshots below for visual example of how I created them:



The screenshot shows the AWS Management Console interface for creating a new subnet. The breadcrumb navigation at the top indicates the path: VPC > Subnets > Create subnet. The main heading is "Create subnet" with an "info" link. The form is divided into two main sections: "VPC" and "Subnet settings".

VPC Section:

- VPC ID:** A dropdown menu with the text "Create subnets in this VPC." above it. The selected value is "vpc-031c57b95dfed4773 (multi-tier-vpc)".
- Associated VPC CIDRs:** A table with one entry:

IPv4 CIDRs
10.10.0.0/16

Subnet settings Section:

- Subnet settings:** A heading with the instruction "Specify the CIDR blocks and Availability Zone for the subnet."
- Subnet 1 of 6:** A label indicating the current step in the process.

Subnet 1 of 6

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

IPv4 subnet CIDR block

256 IPs

< > ^ v

▼ Tags - optional

Key

×

Value - optional

×

Remove

Add new tag

You can add 49 more tags.

Remove

Subnet 2 of 6

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

IPv4 subnet CIDR block

256 IPs

< > ^ v

▼ Tags - optional

Key

×

Value - optional

×

Remove

Add new tag

You can add 49 more tags.

Remove

Subnet 3 of 6

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability Zone [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

IPv4 VPC CIDR block [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

IPv4 subnet CIDR block

256 IPs

< > ^ v

▼ Tags - optional

Key

×

Value - optional

×

Remove

Add new tag

You can add 49 more tags.

Remove

Subnet 4 of 6

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

app-tier-private-2

The name can be up to 256 characters long.

Availability Zone

Info

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (N. Virginia) / us-east-1b

IPv4 VPC CIDR block

Info

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.10.0.0/16

IPv4 subnet CIDR block

10.10.4.0/24

256 IPs

< > ^ v

Tags - optional

Key

Q Name

X

Value - optional

Q app-tier-private-2

X

Remove

Add new tag

You can add 49 more tags.

Remove

Subnet 5 of 6

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

database-tier-private-1

The name can be up to 256 characters long.

Availability Zone

Info

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (N. Virginia) / us-east-1a

IPv4 VPC CIDR block

Info

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.10.0.0/16

IPv4 subnet CIDR block

10.10.5.0/24

256 IPs

< > ^ v

Tags - optional

Key

Q Name

X

Value - optional

Q database-tier-private-1

X

Remove

Add new tag

You can add 49 more tags.

Remove

Subnet 6 of 6

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

database-tier-private-2

The name can be up to 256 characters long.

Availability Zone

Info

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

United States (N. Virginia) / us-east-1b

IPv4 VPC CIDR block

Info

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.10.0.0/16

IPv4 subnet CIDR block

10.10.6.0/24

256 IPs

< > ^ v

Tags - optional

Key

Q Name

X

Value - optional

Q database-tier-private-2

X

Remove

Add new tag

You can add 49 more tags.

Remove

Add new subnet

Cancel

Create subnet

We can see below that all 6 subnets have been created.

VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

- Your VPCs
- Subnets**
- Route tables
- Internet gateways
- Egress-only internet gateways
- Carrier gateways
- DHCP option sets

Subnets (6/12) Info

Find resources by attribute or tag

Name	Subnet ID	State	VPC	Block Public...	IPv4 CID
database-tier-private-2	subnet-0d3d4fe1ea17d3948	Available	vpc-031c57b95dfed4773 mult...	Off	10.10.6.0/24
app-tier-private-2	subnet-0441fa75497cf7de9	Available	vpc-031c57b95dfed4773 mult...	Off	10.10.4.0/24
database-tier-private-1	subnet-01d44f1323f1bccf5	Available	vpc-031c57b95dfed4773 mult...	Off	10.10.5.0/24
web-tier-public-2	subnet-0b33cca6c2ad5848e	Available	vpc-031c57b95dfed4773 mult...	Off	10.10.2.0/24
web-tier-public-1	subnet-02048a2877a8501c9	Available	vpc-031c57b95dfed4773 mult...	Off	10.10.1.0/24
app-tier-private-1	subnet-03082b07ed85f59e6	Available	vpc-031c57b95dfed4773 mult...	Off	10.10.3.0/24

Subnets: subnet-03082b07ed85f59e6, subnet-02048a2877a8501c9, subnet-0b33cca6c2ad5848e, subnet-01d44f1323f1bccf5, subnet-0441fa75497cf7de9, subnet-0d3d4fe1ea17d3948

--Lets set "Enable auto-assign public IPv4 address" in the public subnet settings. Just select the one subnet at a time, go to "Actions" on the top, and go to "Edit subnet settings" and Save it.

-Do the same for all 6 subnets in the VPC.

VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

- Your VPCs
- Subnets**
- Route tables
- Internet gateways
- Egress-only internet gateways
- Carrier gateways

Subnets (1/6) Info

Find resources by attribute or tag

Subnet: vpc-031c57b95dfed4773

Clear filters

Name	Subnet ID	State	VPC
database-tier-private-2	subnet-0d3d4fe1ea17d3948	Available	vpc-031c57b95dfed4773
app-tier-private-2	subnet-0441fa75497cf7de9	Available	vpc-031c57b95dfed4773
database-tier-private-1	subnet-01d44f1323f1bccf5	Available	vpc-031c57b95dfed4773
web-tier-public-2	subnet-0b33cca6c2ad5848e	Available	vpc-031c57b95dfed4773
web-tier-public-1	subnet-02048a2877a8501c9	Available	vpc-031c57b95dfed4773
app-tier-private-1	subnet-03082b07ed85f59e6	Available	vpc-031c57b95dfed4773

Actions

- View details
- Create flow log
- Edit subnet settings**
- Edit IPv6 CIDRs
- Edit network ACL association
- Edit route table association
- Edit CIDR reservations
- Share subnet
- Manage tags
- Delete subnet

Edit subnet settings Info

Subnet

Subnet ID: subnet-02048a2877a8501c9

Name: web-tier-public-1

Auto-assign IP settings Info

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ Enable auto-assign public IPv4 address Info

☐ Enable auto-assign customer-owned IPv4 address Info

Option disabled because no customer owned pools found.

Resource-based name (RBN) settings Info

Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.

☐ Enable resource name DNS A record on launch Info

☐ Enable resource name DNS AAAA record on launch Info

Hostname type Info

☐ Resource name

☒ IP name

DNS64 settings

Enable DNS64 to allow IPv6-only services in Amazon VPC to communicate with IPv4-only services and networks.

☒ Enable DNS64 Info

Cancel Save

Step 3: Internet gateway

Let's navigate to Internet gateway section on the VPC dashboard and then click on "create internet gateway".

Give name: multi-tier-project-IGW and click on Create internet gateway

The screenshot shows the 'Create internet gateway' page in the AWS Management Console. The breadcrumb navigation is 'VPC > Internet gateways > Create internet gateway'. The page title is 'Create internet gateway' with an 'Info' link. A description states: 'An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.' The 'Internet gateway settings' section includes a 'Name tag' field with the value 'multi-tier-project-IGW'. The 'Tags - optional' section shows a table with one tag: Key 'Name', Value 'multi-tier-project-IGW'. At the bottom right are 'Cancel' and 'Create internet gateway' buttons.

Create internet gateway [Info](#)

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

multi-tier-project-IGW

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Name	multi-tier-project-IGW	Remove

[Add new tag](#)
You can add 49 more tags.

[Cancel](#) [Create internet gateway](#)

-- Let's make sure we attach it to the VPC we just made. To do that, go to "actions" and select "attach to VPC".

The screenshot shows the 'Internet gateway details' page for 'igw-0b504023c8d4181d3 / multi-tier-project-IGW'. The breadcrumb navigation is 'VPC > Internet gateways > igw-0b504023c8d4181d3'. The page title is 'igw-0b504023c8d4181d3 / multi-tier-project-IGW'. The 'Details' section shows: Internet gateway ID 'igw-0b504023c8d4181d3', State 'Detached', VPC ID '-', and Owner '284514300790'. The 'Tags' section shows a table with one tag: Key 'Name', Value 'multi-tier-project-IGW'. The 'Actions' menu is open, showing options: 'Attach to VPC', 'Detach from VPC', 'Manage tags', and 'Delete'. At the bottom right are 'Cancel' and 'Attach internet gateway' buttons.

igw-0b504023c8d4181d3 / multi-tier-project-IGW

Details [Info](#)

Internet gateway ID	State	VPC ID	Owner
igw-0b504023c8d4181d3	Detached	-	284514300790

Tags

Key	Value
Name	multi-tier-project-IGW

Actions

- Attach to VPC
- Detach from VPC
- Manage tags
- Delete

[Manage tags](#)

[Cancel](#) [Attach internet gateway](#)

-- On the next page select the VPC and click on "attach internet gateway".

The screenshot shows the 'Attach to VPC' page in the AWS Management Console. The breadcrumb navigation is 'VPC > Internet gateways > Attach to VPC (igw-0b504023c8d4181d3)'. The page title is 'Attach to VPC (igw-0b504023c8d4181d3)' with an 'Info' link. The 'VPC' section includes a description: 'Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.' The 'Available VPCs' section shows a search bar with the value 'vpc-031c57b95dfed4773'. At the bottom right are 'Cancel' and 'Attach internet gateway' buttons.

Attach to VPC (igw-0b504023c8d4181d3) [Info](#)

VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
Attach the internet gateway to this VPC.

vpc-031c57b95dfed4773

[AWS Command Line Interface command](#)

[Cancel](#) [Attach internet gateway](#)

The VPC is now attached.

VPC > Internet gateways > igw-0b504023c8d4181d3

VPC dashboard < **igw-0b504023c8d4181d3 / multi-tier-project-IGW** Actions

EC2 Global View

Filter by VPC

▼ **Virtual private cloud**

- Your VPCs
- Subnets
- Route tables
- Internet gateways**
- Egress-only internet gateways

Details Info

Internet gateway ID igw-0b504023c8d4181d3

State Attached

VPC ID vpc-031c57b95dfed4773 | multi-tier-vpc

Owner 284514300790

Tags Manage tags

Q Search tags

Key	Value
Name	multi-tier-project-IGW

Step 4: NAT gateway

Let's go to NAT gateway section in the VPC dashboard and click on "Create NAT gateway" to set it up.

aws Search [Alt+S] United States (N. Virginia) srinadhgdr

VPC dashboard < **NAT gateways** Info Actions Create NAT gateway

EC2 Global View

Filter by VPC

▼ **Virtual private cloud**

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways**
- Peering connections

Find resources by attribute or tag

Name	NAT gateway ID	Connectivity...	State	State message	Primary public I...	Primary priva
No NAT gateways found						

Select a NAT gateway

Let's provide below details

Name: multi-tier-project-NATgateway

Subnet: select one of the public subnets(web-tier-public-1)

Elastic IP allocation ID: Click on Allocate Elastic IP and then click "Create NAT gateway" at the bottom.

aws [Search] [Alt+S] United States (N. Virginia) srinadhgdlr

VPC > NAT gateways > Create NAT gateway

A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
multi-tier-project-NATgateway
The name can be up to 256 characters long.

Subnet
Select a subnet in which to create the NAT gateway.
subnet-02048a2877a8501c9 (web-tier-public-1)

Connectivity type
Select a connectivity type for the NAT gateway.
☒ Public
☐ Private

Elastic IP allocation ID [Info](#)
Assign an Elastic IP address to the NAT gateway.
elpallo-0d21d9fbd2b6f894 [Allocate Elastic IP](#)

► **Additional settings** [info](#)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key
Q, Name X

Value - optional
Q, multi-tier-project-NATgateway X [Remove](#)

[Add new tag](#)
You can add 49 more tags.

[Cancel](#) [Create NAT gateway](#)

VPC > NAT gateways > nat-029e7f16a0d60e6db

VPC dashboard < EC2 Global View [Filter by VPC](#)

▼ **Virtual private cloud**
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways

nat-029e7f16a0d60e6db / multi-tier-project-NATgateway

[Actions](#)

Details

NAT gateway ID
nat-029e7f16a0d60e6db

NAT gateway ARN
arn:aws:ec2:us-east-1:284514300790:natgateway/nat-029e7f16a0d60e6db

VPC
vpc-031c57b95dfed4773 / multi-tier-vpc

Connectivity type
Public

Primary public IPv4 address
-

Subnet
subnet-02048a2877a8501c9 / web-tier-public-1

State
Pending

Primary private IPv4 address
-

Created
Friday, April 11, 2025 at 11:10:50 GMT+1

State message [Info](#)
-

Primary network interface ID
-

Deleted
-

Step 5: Route tables

I'll be setting up a public and private route table. Go to Route tables on the VPC dashboard and click on "Create route table".

aws [Search] [Alt+S] United States (N. Virginia) srinadhgdlr

VPC dashboard < EC2 Global View [Filter by VPC](#)

▼ **Virtual private cloud**
Your VPCs
Subnets
Route tables
Internet gateways

Route tables (2) [Info](#)

Find resources by attribute or tag

Last updated 26 minutes ago [Actions](#) [Create route table](#)

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	-	rtb-01d0f52128038d7af	-	-	Yes	vpc-0f982f9178eb91a4f
☐	-	rtb-0e985f3831f69720a	-	-	Yes	vpc-031c57b95dfed4773

1. Create the public route table

Let's provide below details

Name: multi-tier-project-public-route-table

VPC: select our VPC (multi-tier-vpc) and then click “Create route table”.

VPC > Route tables > Create route table

Create route table [Info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

multi-tier-project-public-route-table

VPC
The VPC to use for this route table.

vpc-031c57b95dfed4773 (multi-tier-vpc)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name X

Value - optional

Q multi-tier-project-public-route-table X Remove

Add new tag

You can add 49 more tags.

Cancel Create route table

Now we need to add the two public subnets to it. To do that, we go to the Explicit subnet associations section and click on “Edit subnet associations”.

VPC > Route tables > rtb-0cac71b6f2aefff1c

rtb-0cac71b6f2aefff1c / multi-tier-project-public-route-table [Actions](#)

Details [Info](#)

Route table ID rtb-0cac71b6f2aefff1c	Main No	Explicit subnet associations -	Edge associations -
VPC vpc-031c57b95dfed4773 multi-tier-vpc	Owner ID 284514300790		

Routes Subnet associations Edge associations Route propagation Tags

Explicit subnet associations (0) [Edit subnet associations](#)

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
No subnet associations You do not have any subnet associations.			

Select the two public subnets and click “Save associations”.

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/6)

☐

database-tier-private-2

subnet-0d3d4fe1ea17d3948

10.10.6.0/24

-

Main (rtb-0e985f3831f69720a)

☐

app-tier-private-2

subnet-0441fa75497cf7de9

10.10.4.0/24

-

Main (rtb-0e985f3831f69720a)

☐

database-tier-private-1

subnet-01d44f1323f1bccf5

10.10.5.0/24

-

Main (rtb-0e985f3831f69720a)

☒

web-tier-public-2

subnet-0b33cca6c2ad5848e

10.10.2.0/24

-

Main (rtb-0e985f3831f69720a)

☒

web-tier-public-1

subnet-02048a2877a8501c9

10.10.1.0/24

-

Main (rtb-0e985f3831f69720a)

☐

app-tier-private-1

subnet-03082b07ed85f59e6

10.10.3.0/24

-

Main (rtb-0e985f3831f69720a)

Selected subnetssubnet-02048a2877a8501c9 / web-tier-public-1 subnet-0b33cca6c2ad5848e / web-tier-public-2 [Cancel](#) [Save associations](#)

You have successfully updated subnet associations for rtb-0cac71b6f2aefff1c / multi-tier-project-public-route-table.

rtb-0cac71b6f2aefff1c / multi-tier-project-public-route-table

[Actions](#)

Details [Info](#)

Route table ID
[rtb-0cac71b6f2aefff1c](#)

VPC
[vpc-031c57b95dfed4773](#) | multi-tier-vpc

Main
☐ No

Owner ID
[284514300790](#)

Explicit subnet associations
[2 subnets](#)

Edge associations
-

[Routes](#) [Subnet associations](#) [Edge associations](#) [Route propagation](#) [Tags](#)

Explicit subnet associations (2)

[Edit subnet associations](#)

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
web-tier-public-2	subnet-0b33cca6c2ad5848e	10.10.2.0/24	-
web-tier-public-1	subnet-02048a2877a8501c9	10.10.1.0/24	-

In the same public route table, now we go to Routes and click “Edit routes” to add the internet gateway to the Route.

rtb-0cac71b6f2aefff1c / multi-tier-project-public-route-table

[Actions](#)

Details [Info](#)

Route table ID
[rtb-0cac71b6f2aefff1c](#)

VPC
[vpc-031c57b95dfed4773](#) | multi-tier-vpc

Main
☐ No

Owner ID
[284514300790](#)

Explicit subnet associations
[2 subnets](#)

Edge associations
-

[Routes](#) [Subnet associations](#) [Edge associations](#) [Route propagation](#) [Tags](#)

Routes (1)

[Both](#) [Edit routes](#)

Destination	Target	Status	Propagated
10.10.0.0/16	local	Active	No

Click "Add route" and select 0.0.0.0/0, then "Internet Gateway" for the target. Click "Save changes".

VPC > Route tables > rtb-0cac71b6f2aefff1c > Edit routes

Edit routes

Destination	Target	Status	Propagated
10.10.0.0/16	local	Active	No
0.0.0.0/0	Internet Gateway	-	No

Use: "igw-0b504023c8d4181d3"
 igw-0b504023c8d4181d3 (multi-tier-project-IGW)

VPC > Route tables > rtb-0cac71b6f2aefff1c

rtb-0cac71b6f2aefff1c / multi-tier-project-public-route-table

Actions

Details Info

Route table ID rtb-0cac71b6f2aefff1c	Main No	Explicit subnet associations 2 subnets	Edge associations -
VPC vpc-031c57b95dfed4773 multi-tier-vpc	Owner ID 284514300790		

[Routes](#) | [Subnet associations](#) | [Edge associations](#) | [Route propagation](#) | [Tags](#)

Routes (2)

Filter routes

Destination	Target	Status	Propagated
0.0.0.0/0	igw-0b504023c8d4181d3	Active	No
10.10.0.0/16	local	Active	No

2. Create the private route table

Now create the private route table. We can just repeat the above process of naming the table and selecting the VPC but we should include private on the name so we can find it later.

Let's provide below details

Name: multi-tier-project-private-route-table

VPC: select our VPC (multi-tier-vpc) and then click "Create route table".

aws | Search [Alt+S] | United States (N. Virginia) | srinadgdlr

VPC > Route tables > Create route table

Create route table

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
 Create a tag with a key of 'Name' and a value that you specify.
 multi-tier-project-private-route-table

VPC
 The VPC to use for this route table.
 vpc-031c57b95dfed4773 (multi-tier-vpc)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key
 Name

Value - optional
 multi-tier-project-private-route-table

You can add 49 more tags.

When we go to “Edit subnet associations” of Subnet associations, we’re going to check all 4 private subnets and click “Save associations”.

VPC > Route tables > rtb-0bb223565a23d92dc > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (4/6)

Filter subnet associations

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	database-tier-private-2	subnet-0d3d4fe1ea17d3948	10.10.6.0/24	-	Main (rtb-0e985f3831f69720a)
☒	app-tier-private-2	subnet-0441fa75497cf7de9	10.10.4.0/24	-	Main (rtb-0e985f3831f69720a)
☒	database-tier-private-1	subnet-01d44f1323f1bccf5	10.10.5.0/24	-	Main (rtb-0e985f3831f69720a)
☐	web-tier-public-2	subnet-0b33ca6c2ad5848e	10.10.2.0/24	-	rtb-0cac71b6f2aefff1c / multi-tier-proj...
☐	web-tier-public-1	subnet-02048a2877a8501c9	10.10.1.0/24	-	rtb-0cac71b6f2aefff1c / multi-tier-proj...
☒	app-tier-private-1	subnet-03082b07ed85f59e6	10.10.3.0/24	-	Main (rtb-0e985f3831f69720a)

Selected subnets

subnet-0d3d4fe1ea17d3948 / database-tier-private-2

subnet-0441fa75497cf7de9 / app-tier-private-2

subnet-01d44f1323f1bccf5 / database-tier-private-1

subnet-03082b07ed85f59e6 / app-tier-private-1

Cancel

Save associations

VPC > Route tables > rtb-0bb223565a23d92dc

rtb-0bb223565a23d92dc / multi-tier-project-private-route-table

Actions

VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Details

Route table ID

rtb-0bb223565a23d92dc

VPC

vpc-031c57b95dfed4773 | multi-tier-vpc

Main

No

Owner ID

284514300790

Explicit subnet associations

4 subnets

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Explicit subnet associations (4)

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
database-tier-private-2	subnet-0d3d4fe1ea17d3948	10.10.6.0/24	-
app-tier-private-2	subnet-0441fa75497cf7de9	10.10.4.0/24	-
database-tier-private-1	subnet-01d44f1323f1bccf5	10.10.5.0/24	-
app-tier-private-1	subnet-03082b07ed85f59e6	10.10.3.0/24	-

Edit subnet associations

For the Routes, we’re going to add a route with destination 0.0.0.0/0 on this private route table too but for the target, we have to select NAT Gateway and Save changes.

VPC > Route tables > rtb-0bb223565a23d92dc

rtb-0bb223565a23d92dc / multi-tier-project-private-route-table

Actions

VPC dashboard

EC2 Global View

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Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

Details

Route table ID

rtb-0bb223565a23d92dc

VPC

vpc-031c57b95dfed4773 | multi-tier-vpc

Main

No

Owner ID

284514300790

Explicit subnet associations

4 subnets

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (1)

Filter routes

Destination	Target	Status	Propagated
10.10.0.0/16	local	Active	No

Both

Edit routes

VPC > Route tables > rtb-0bb223565a23d92dc > Edit routes

Edit routes

Destination	Target	Status	Propagated
10.10.0.0/16	local	Active	No
0.0.0.0/0	NAT Gateway	-	No

Use: "nat-029e7f16a0d60e6db"
 nat-029e7f16a0d60e6db (multi-tier-project-NATgateway)

VPC > Route tables > rtb-0bb223565a23d92dc

rtb-0bb223565a23d92dc / multi-tier-project-private-route-table

Details Info

Route table ID
rtb-0bb223565a23d92dc

VPC
vpc-031c57b95dfed4773 | multi-tier-vpc

Main

No

Owner ID

284514300790

Explicit subnet associations

4 subnets

Edge associations

-

[Routes](#) | Subnet associations | Edge associations | Route propagation | Tags

Routes (2)

Destination	Target	Status	Propagated
0.0.0.0/0	nat-029e7f16a0d60e6db	Active	No
10.10.0.0/16	local	Active	No

Step 6: Create Security groups

Let's go to Security groups section in the VPC dashboard and click on "Create security group" to set up.

aws [Alt+S] United States (N. Virginia) srinadhgdlr

VPC dashboard < EC2 Global View

Virtual private cloud

Security

Network ACLs

Security groups

PrivateLink and

Security Groups (2) Info

☐	Name	Security group ID	Security group name	VPC ID	Description
☐	-	sg-04fe31b763f20b828	default	vpc-031c57b95dfed4773	default VPC s
☐	-	sg-0b4024219fbf57905	default	vpc-0f982f9178eb91a4f	default VPC s

Provide below details in the Security group section

Security group name: multi-tier-project-securitygroup

Description: Allow SSH access

VPC: choose multi-tier-vpc

Inbound rules::

Click "Add rule".

Select ssh for the Type section, and 0.0.0.0/0 for the source. Click “Add rule”.

Select HTTP for the type and 0.0.0.0/0 for the source.

Leave the rest of things

Click on Create security group at the bottom.

Search

[Alt+S]

United States (N. Virginia)srinadhgdr

VPC > Security Groups > Create security group

Create security group

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name

multi-tier-project-securitygroup

Name cannot be edited after creation.

Description

Allow SSH access

VPC

vpc-031c57b95dfed4773 (multi-tier-vpc)

Inbound rules

Type	Protocol	Port range	Source	Description - optional
SSH	TCP	22	Anywh...	
			0.0.0.0/0	
HTTP	TCP	80	Anywh...	
			0.0.0.0/0	

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Outbound rules

Type	Protocol	Port range	Destination	Description - optional
All traffic	All	All	Custom	
			0.0.0.0/0	

Add rule

Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags

Cancel

Create security group

sg-0ae1b5fa1429f71e3 - multi-tier-project-securitygroup

Actions ▾

Details

Security group name
 multi-tier-project-securitygroupSecurity group ID
 sg-0ae1b5fa1429f71e3Description
 Allow SSH accessVPC ID
 vpc-031c57b95dfed4773Owner
 284514300790Inbound rules count
2 Permission entriesOutbound rules count
1 Permission entry[Inbound rules](#) | [Outbound rules](#) | [Sharing - new](#) | [VPC associations - new](#) | [Tags](#)

Inbound rules (2)



Manage tags

[Edit inbound rules](#)< 1 > 

☐	Name ▾	Security group rule ID ▾	IP version ▾	Type ▾	Protocol ▾	Port range ▾	Source
☐	-	sgr-00a10559d60e2f9cc	IPv4	SSH	TCP	22	0.0.0.0/0
☐	-	sgr-08bbc459ea93399ec1	IPv4	HTTP	TCP	80	0.0.0.0/0

Create app-tier-securitygroup for APP Tier

Let's go to Security groups section in the VPC dashboard and click on "Create security group" to set up.

Provide below details in the Security group section

Security group name: app-tier-securitygroup

Description: Allow SSH access

VPC: choose multi-tier-vpc

Inbound rules::

Click "Add rule".

Select ssh for type, and select the security group multi-tier-project-securitygroup as the source.

Click "Add rule".

select "All ICMP-IPv4" for the type, and 0.0.0.0/0 as the source.

Leave the rest of things

Click on Create security group at the bottom.

aws [Search] [Alt+S] United States (N. Virginia) srinadhgdlr

VPC > Security Groups > Create security group

Create security group [info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [info](#)

Name cannot be edited after creation.

Description [info](#)

VPC [info](#)

Inbound rules [info](#)

Type info	Protocol info	Port range info	Source info	Description - optional info	
SSH	TCP	22	Custom sg-0ae1b5fa1429f71e3		Delete
All ICMP - IPv4	ICMP	All	Anyw... 0.0.0.0/0		Delete

VPC > Security Groups > sg-0da0117844fbcd3c2 - app-tier-securitygroup

sg-0da0117844fbcd3c2 - app-tier-securitygroup [Actions](#)

Details

Security group name app-tier-securitygroup	Security group ID sg-0da0117844fbcd3c2	Description Allow SSH access	VPC ID vpc-031c57b95dfed4773
Owner 284514300790	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

[Inbound rules](#) | [Outbound rules](#) | [Sharing - new](#) | [VPC associations - new](#) | [Tags](#)

Inbound rules (2)

[Manage tags](#) [Edit inbound rules](#)

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
☐	-	sgr-0b01981ea37c8f780	IPv4	All ICMP - IPv4	ICMP	All	0.0.0.0/0	-
☐	-	sgr-059ac95eabe7bfe84	-	SSH	TCP	22	sg-0ae1b5fa1429f71e3 / multi-tier-project-securitygroup	-

Create database-tier-securitygroup for database Tier

Let's go to Security groups section in the VPC dashboard and click on "Create security group" to set up.

Provide below details in the Security group section

Security group name: database-tier-securitygroup

Description: Allow SSH access

VPC: choose multi-tier-vpc

Inbound rules::

Click "Add rule".

Select ssh for type, and select the security group multi-tier-project-securitygroup as the source.

Click "Add rule".

select "All ICMP-IPv4" for the type, and 0.0.0.0/0 as the source.

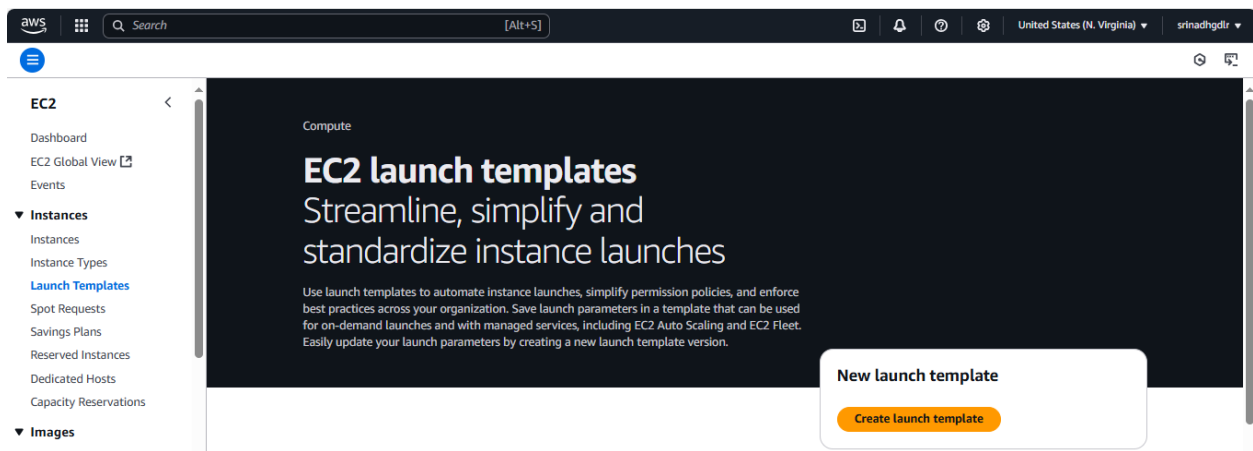
Leave the rest of things

Click on Create security group at the bottom.

Now I have the infrastructure needed to set up the three tiers. I am going to build the tiers one at a time starting with the web tier.

Web Tier

Go to Launch templates in the EC2 dashboard and click on “Create launch template”.



Let's provide below details

Launch template name: web-tier-template

Template version description: web-tier-template

Auto Scaling guidance: check the box "Provide guidance to help me set up a template that I can use with EC2 Auto Scaling"

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - *required*

web-tier-template

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '\', '@'.

Template version description

web-tier-template

Max 255 chars

Auto Scaling guidance | Info

Select this if you intend to use this template with EC2 Auto Scaling

☒ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

► Template tags

► Source template

Launch template contents

On the next Launch template contents section below, choose a free tier OS. I go with Amazon Linux.

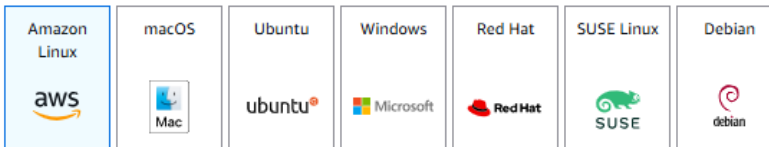
▼ Application and OS Images (Amazon Machine Image) - required Info

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

🔍 Search our full catalog including 1000s of application and OS images

Recents

Quick Start



🔍
[Browse more AMIs](#)
Including AMIs from
AWS, Marketplace and
the Community

Amazon Machine Image (AMI)

Amazon Linux 2 AMI (HVM) - Kernel 5.10, SSD Volume Type
ami-07a6f770277670015 (64-bit (x86)) / ami-06357f5cb4aea07d8 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

Description

Amazon Linux 2 comes with five years support. It provides Linux kernel 5.10 tuned for optimal performance on Amazon EC2, systemd 219, GCC 7.3, Glibc 2.26, Binutils 2.29.1, and the latest software packages through extras. This AMI is the successor of the Amazon Linux AMI that is now under maintenance only mode and has been removed from this wizard.

Amazon Linux 2 Kernel 5.10 AMI 2.0.20250405.0 x86_64 HVM gp2

Architecture

64-bit (x86) ▼

AMI ID

ami-07a6f770277670015

Publish Date

2025-04-05

Username | ⓘ

ec2-user

Verified provider

Instance type: t2.micro

Key pair (login): multi-tier-project-keypair

(I created a new key pair multi-tier-project-keypair for this project.)

▼ Instance type [Info](#) | [Get advice](#)

Advanced

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Windows base pricing: 0.0162 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour

On-Demand SUSE base pricing: 0.0116 USD per Hour On-Demand RHEL base pricing: 0.026 USD per Hour

On-Demand Linux base pricing: 0.0116 USD per Hour

Free tier eligible

☒ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

multi-tier-project-keypair ▼

[Create new key pair](#)

For the Network settings section

Firewall (security groups): select existing security group - multi-tier-project-securitygroup

In the Advanced network configuration section, click “add network interface”.

Select Enable for Auto-assign public IP.

▼ Network settings Info

Subnet Info

Don't include in launch template ▼

Create new subnet ↗

When you specify a subnet, a network interface is automatically added to your template.

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Select existing security group

☐ Create security group

Common security groups Info

Select security groups ▼

multi-tier-project-securitygroup sg-0ae1b5fa1429f71e3 X
VPC: vpc-031c57b95dfed4773

Compare security group rules ↻

Security groups that you add or remove here will be added to or removed from all your network interfaces.

▼ Advanced network configuration

Network interface 1

Device index Info

0

Network interface Info

New interface ▼

Description Info

Remove

Existing network interfaces are not recommended when creating a template for auto-scaling.

Subnet Info

Don't include in launch template

Not applicable for EC2 Auto Scaling

Security groups Info

Select security groups ▼

Auto-assign public IP Info

Enable ▼

▼ Storage (volumes) Info

EBS Volumes

Hide details

► Volume 1 (AMI Root) : 8 GiB, EBS, General purpose SSD (gp2)

AMI Volumes are not included in the template unless modified

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage X

Add new volume

▼ Resource tags Info

No resource tags are currently included in this template. Add a resource tag to include it in the launch template.

Add new tag

You can add up to 50 more tags.

► Advanced details Info

Now let's go to Advanced details

▼
Advanced details
Info

IAM instance profile
Info

EC2InstanceProfile
arn:aws:iam::284514300790:instance-profile/EC2InstanceProfile

Create new IAM profile

Hostname type
Info

Don't include in launch template

and now scroll down to User data. We're going to enter a script here for an Apache web server.

```
#!/bin/bash
```

```
#Update
```

```
sudo yum update -y
```

```
#Install
```

```
sudo yum install -y httpd
```

```
#Start
```

```
sudo systemctl start httpd
```

```
#Enable
```

```
sudo systemctl enable httpd
```

```
#Status
```

```
sudo systemctl status httpd
```

User data - optional
Info

Upload a file with your user data or enter it in the field.

Choose file

```
#!/bin/bash
#Update
sudo yum update -y

#Install
sudo yum install -y httpd

#Start
sudo systemctl start httpd

#Enable
sudo systemctl enable httpd

#Status
sudo systemctl status httpd
```

☐ User data has already been base64 encoded

Virtual server type (instance type)
t2.micro

Firewall (security group)
multi-tier-project-securitygroup

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Cancel
Create launch template

Lastly, click "Create launch template".

EC2 > Launch Templates > Create launch template

Success
Successfully created `web-tier-template(it-0a3db827d5495d1ce)`.

► Actions log

Next Steps

Launch an instance
With On-Demand Instances, you pay for compute capacity by the second (for Linux, with a minimum of 60 seconds) or by the hour (for all other operating systems) with no long-term commitments or upfront payments. Launch an On-Demand Instance from your launch template.
[Launch instance from this template](#)

Create an Auto Scaling group from your template
Amazon EC2 Auto Scaling helps you maintain application availability and allows you to scale your Amazon EC2 capacity up or down automatically according to conditions you define. You can use Auto Scaling to help ensure that you are running your desired number of Amazon EC2 instances during demand spikes to maintain performance and decrease capacity during lulls to reduce costs.
[Create Auto Scaling group](#)

Create Spot Fleet
A Spot Instance is an unused EC2 instance that is available for less than the On-Demand price. Because Spot Instances enable you to request unused EC2 instances at steep discounts, you can lower your Amazon EC2 costs significantly. The hourly price for a Spot Instance (of each instance type in each Availability Zone) is set by Amazon EC2, and adjusted gradually based on the long-term supply of and demand for Spot Instances. Spot instances are well-suited for data-analysis, batch jobs, background processing, and optional tasks.
[Create Spot Fleet](#)

[View launch templates](#)

So far so good!

Before we try to create the next tier, let's **Create Auto Scaling group**.

Go to Launch Templates in the EC2 dashboard and select the Launch Template Name (`web-tier-template`), then click on Actions dropdown button and select "Create Auto Scaling group".

aws Search [Alt+S] United States (N. Virginia) srinadhgdlr

EC2

Dashboard
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Events

▼ Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations

▼ Images

Launch Templates (1/1) Info

Search

Launch Template ID	Launch Template Name	Default Version
it-0a3db827d5495d1ce	web-tier-template	1

Actions

- Launch instance from template
- Modify template (Create new version)
- Delete template
- Delete template version
- Set default version
- Manage tags
- Create Spot Fleet
- Create Auto Scaling group**
- View details

web-tier-template (it-0a3db827d5495d1ce)

Launch template details

Launch template ID it-0a3db827d5495d1ce	Launch template name web-tier-template	Default version 1	Owner arn:aws:iam::284514300790:root
---	--	-----------------------------	--

Actions Delete template

Provide below details Name

Auto Scaling group name: multi-tier-auto-scaling-group

Launch template: web-tier-template

Click on Next

Step 1

● Choose launch template

○ Step 2

○ Step 3 - optional

○ Step 4 - optional

○ Step 5 - optional

○ Step 6 - optional

○ Step 7

Choose launch template [Info](#)

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.

Name
Auto Scaling group name
Enter a name to identify the group.

Must be unique to this account in the current Region and no more than 255 characters.

Launch template [Info](#)

① For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template
Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.
 [Create a launch template](#) [Info](#)

Version
 [Create a launch template version](#) [Info](#)

Create a launch template version [Info](#)

Description web-tier-template	Launch template web-tier-template Info lt-0a3db827d5495d1ce	Instance type t2.micro
AMI ID ami-07a6f770277670015	Security groups -	Request Spot Instances No
Key pair name multi-tier-project-keypair	Security group IDs sg-0ae1b5fa1429f71e3 Info	

Additional details

Storage (volumes) -	Date created Fri Apr 11 2025 12:46:17 GMT+0100 (British Summer Time)
-------------------------------	--

[Cancel](#) [Next](#)

Select the VPC for this project and the two public web tier subnets.
Click "Next"

Step 1

● Choose launch template

○ Step 2

● Choose instance launch options

○ Step 3 - optional

○ Step 4 - optional

○ Step 5 - optional

○ Step 6 - optional

○ Step 7

Choose instance launch options [Info](#)

Choose the VPC network environment that your instances are launched into, and customize the instance types and purchase options.

Instance type requirements [Info](#)
You can keep the same instance attributes or instance type from your launch template, or you can choose to override the launch template by specifying different instance attributes or manually adding instance types.

Launch template web-tier-template Info lt-0a3db827d5495d1ce	Version Default	Description web-tier-template
--	---------------------------	---

Instance type
t2.micro

Network [Info](#)
For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.
VPC
Choose the VPC that defines the virtual network for your Auto Scaling group.
 [Info](#)

[Override launch template](#)

[Create a VPC](#)

Availability Zones and subnets
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

us-east-1a | subnet-0b33cca6c2ad5848e (web-tier-public-2) ✕
10.10.2.0/24

us-east-1a | subnet-02048a2877a8501c9 (web-tier-public-1) ✕
10.10.1.0/24

[Create a subnet](#)

Availability Zone distribution - new
Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

☒ **Balanced best effort**
If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.

☐ **Balanced only**
If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy Availability Zone to preserve balanced distribution.

Cancel [Skip to review](#) [Previous](#) [Next](#)

Select Load balancing: Attach to a new load balancer
 Load balancer type: Application load balancer
 Load balancer name: multi-tier-auto-scaling-group-1
 Load balancer scheme: Internet-facing.
 Listeners and routing: select Create a target group and click "Next".

[EC2](#) > [Auto Scaling groups](#) > Create Auto Scaling group

①

Step 1
● Choose launch template

Step 2
● Choose instance launch options

Step 3 - optional
● **Integrate with other services**

Step 4 - optional
● Configure group size and scaling

Step 5 - optional
● Add notifications

Step 6 - optional
● Add tags

Step 7
● Review

Integrate with other services - optional [Info](#)
Use a load balancer to distribute network traffic across multiple servers. Enable service-to-service communications with VPC Lattice. Shift resources away from impaired Availability Zones with zonal shift. You can also customize health check replacements and monitoring.

Load balancing [Info](#)
Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

☐ No load balancer
Traffic to your Auto Scaling group will not be fronted by a load balancer.

☐ Attach to an existing load balancer
Choose from your existing load balancers.

☒ **Attach to a new load balancer**
Quickly create a basic load balancer to attach to your Auto Scaling group.

Attach to a new load balancer
Define a new load balancer to create for attachment to this Auto Scaling group.

Load balancer type
Choose from the load balancer types offered below. Type selection cannot be changed after the load balancer is created. If you need a different type of load balancer than those offered here, visit the [Load Balancing console](#). [Info](#)

☒ **Application Load Balancer**
HTTP, HTTPS

☐ Network Load Balancer
TCP, UDP, TLS

Load balancer name
Name cannot be changed after the load balancer is created.
multi-tier-auto-scaling-group-1

Load balancer scheme
Scheme cannot be changed after the load balancer is created.

☐ Internal

☒ **Internet-facing**

Network mapping
Your new load balancer will be created using the same VPC and Availability Zone selections as your Auto Scaling group. You can select different subnets and add subnets from additional Availability Zones.

VPC
vpc-031c57b95dfed4773 [Info](#) multi-tier-vpc

Availability Zones and subnets
You must select a single subnet for each Availability Zone enabled. Only public subnets are available for selection to support DNS resolution.

☒ us-east-1b subnet-07b933881ec0f1a1d

☒ us-east-1a subnet-02048a2877a8501c9

Listeners and routing
If you require secure listeners, or multiple listeners, you can configure them from the [Load Balancing console](#) [Info](#) after your load balancer is created.

Protocol HTTP

Port 80

Default routing (forward to)
Create a target group

New target group name
An instance target group with default settings will be created.
multi-tier-auto-scaling-group-1

Tags - optional

Health checks

Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.

EC2 health checks
[Always enabled](#)

Additional health check types - optional [Info](#)

☐ **Turn on Elastic Load Balancing health checks** Recommended
Elastic Load Balancing monitors whether instances are available to handle requests. When it reports an unhealthy instance, EC2 Auto Scaling can replace it on its next periodic check.

☐ **Turn on VPC Lattice health checks**
VPC Lattice can monitor whether instances are available to handle requests. If it considers a target as failed a health check, EC2 Auto Scaling replaces it after its next periodic check.

☐ **Turn on Amazon EBS health checks**
EBS monitors whether an instance's root volume or attached volume stalls. When it reports an unhealthy volume, EC2 Auto Scaling can replace the instance on its next periodic health check.

Health check grace period [Info](#)
This time period delays the first health check until your instances finish initializing. It doesn't prevent an instance from terminating when placed into a non-running state.

seconds

[Cancel](#)
[Skip to review](#)
[Previous](#)
[Next](#)

Select 2 for Desired capacity and Min desired capacity, 4 for Max desired capacity. Leave everything else in default and click "Next" and click "Next" again from here to the final screen and click "Create auto scaling group".

[EC2](#) > [Auto Scaling groups](#) > Create Auto Scaling group

Step 1
Choose launch template
Step 2
Choose instance launch options
Step 3 - optional
Integrate with other services
Step 4 - optional
Configure group size and scaling
Step 5 - optional
Add notifications
Step 6 - optional
Add tags
Step 7
Review

Configure group size and scaling - optional

Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size [Info](#)

Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type
Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

Desired capacity
Specify your group size.

Scaling [Info](#)

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits
Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

Equal or less than desired capacity

Max desired capacity

Equal or greater than desired capacity

Automatic scaling - optional
Automatic scaling - optional
Choose whether to use a target tracking policy [Info](#)
You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☒ **No scaling policies**
Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☐ **Target tracking scaling policy**
Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Instance maintenance policy [Info](#)

Control your Auto Scaling group's availability during instance replacement events. This includes health checks, instance refreshes, maximum instance lifetime features and events that happen automatically to keep your group balanced, called rebalancing events.

Choose a replacement behavior depending on your availability requirements

Mixed behavior

☒ **No policy**
For rebalancing events, new instances will launch before terminating others. For all other events, instances terminate and launch at the same time.

Prioritize availability

☐ **Launch before terminating**
Launch new instances and wait for them to be ready before terminating others. This allows you to go above your desired capacity by a given percentage and may temporarily increase costs.

Control costs

☐ **Terminate and launch**
Terminate and launch instances at the same time. This allows you to go below your desired capacity by a given percentage and may temporarily reduce availability.

Flexible

☐ **Custom behavior**
Set custom values for the minimum and maximum amount of available capacity. This gives you greater flexibility in setting how far below and over your desired capacity EC2 Auto Scaling goes when replacing instances.

Additional capacity settings

Capacity Reservation preference [Info](#)

Select whether you want Auto Scaling to launch instances into an existing Capacity Reservation or Capacity Reservation resource group.

- ☒ **Default**
Auto Scaling uses the Capacity Reservation preference from your launch template.
- ☐ **None**
Instances will not be launched into a Capacity Reservation.
- ☐ **Capacity Reservations only**
Instances will only be launched into a Capacity Reservation. If capacity isn't available, the instances fail to launch.
- ☐ **Capacity Reservations first**
Instances will attempt to launch into a Capacity Reservation first. If capacity isn't available, instances will run in On-Demand capacity.

Additional settings

Instance scale-in protection

If protect from scale in is enabled, newly launched instances will be protected from scale in by default.

☐ Enable instance scale-in protection

Monitoring [Info](#)

☐ Enable group metrics collection within CloudWatch

Default instance warmup [Info](#)

The amount of time that CloudWatch metrics for new instances do not contribute to the group's aggregated instance metrics, as their usage data is not reliable yet.

☐ Enable default instance warmup

[Cancel](#)

[Skip to review](#)

[Previous](#)

[Next](#)

Click Next on Add Notifications -optional screen

Click Next on Add tags -optional screen

On the Review tab click on Create Auto Scaling group

Step 1

● Choose launch template

Step 2

● Choose instance launch options

Step 3 - optional

● Integrate with other services

Step 4 - optional

● Configure group size and scaling

Step 5 - optional

● Add notifications

Step 6 - optional

● Add tags

Step 7

● **Review**

Review [Info](#)

Step 1: Choose launch template [Edit](#)

Group details

Auto Scaling group name

multi-tier-auto-scaling-group

Launch template

Launch template

web-tier-template [Info](#)

lt-0a3db827d5495d1ce

Version

Default

Description

web-tier-template

Step 2: Choose instance launch options [Edit](#)

Network

VPC

vpc-031c57b95dfed4773 [Info](#)

Availability Zones and subnets

Availability Zone	Subnet	Subnet CIDR range
us-east-1a	subnet-02048a2877a8501c9 Info	10.10.1.0/24
us-east-1b	subnet-07b933881ec0f1a1d Info	10.10.2.0/24

Availability Zone distribution

Balanced best effort

Instance type requirements

This Auto Scaling group will adhere to the launch template.

Step 3: Integrate with other services

Edit

Load balancing

Load balancer 1

Name	Type	Target group
multi-tier-auto-scaling-group-1	Application/HTTP	multi-tier-auto-scaling-group-1

VPC Lattice integration options

VPC Lattice target groups	
-	

Application Recovery Controller (ARC) zonal shift

ARC zonal shift	
Disabled	

Health checks

Health check type	Health check grace period	
EC2	300 seconds	

Step 4: Configure group size and scaling policies

Edit

Group size

Desired capacity	Desired capacity type	
2	Units (number of instances)	

Scaling

Minimum desired capacity	Maximum desired capacity	
2	4	
Target tracking policy		
-		

Instance maintenance policy

Replacement behavior	Min healthy percentage	Max healthy percentage
No policy	-	-

Additional settings

Instance scale-in protection

Disabled

Monitoring

Disabled

Default instance warmup

Disabled

Capacity Reservation preference

Preference

Default

Capacity Reservation IDs

-

Resource Groups

-

Step 5: Add notifications

Notifications

No notifications

Edit

Step 6: Add tags

Tags (0)

Key	Value	Tag new instances
No tags		

Edit

Preview code

Cancel

Previous

Create Auto Scaling group

aws

Search

[Alt+S]

United States (N. Virginia)

srinadgdlr

EC2

>

Auto Scaling groups

Auto Scaling groups (1)

Launch configurations

Launch templates

Actions

Create Auto Scaling group

Search your Auto Scaling groups

< 1 >

☐	Name	Launch template/configuration	Instances	Status	Desired capacity	Min	Max	Availability Zones
☐	multi-tier-auto-scaling-group	web-tier-template Version Default	0	Updating capacity...	2	2	4	us-east-1a, us-east-1b

We should have two instances running... and here they are.

Instances (1/2)

Last updated 1 minute ago

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

< 1 >

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv
☒		i-045e3976f7d23c3c7	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1b	ec2-44-200-517
☐		i-0be5568922dee2c7e	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	ec2-107-2

i-045e3976f7d23c3c7

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Instance summary

Instance ID

i-045e3976f7d23c3c7

Public IPv4 address

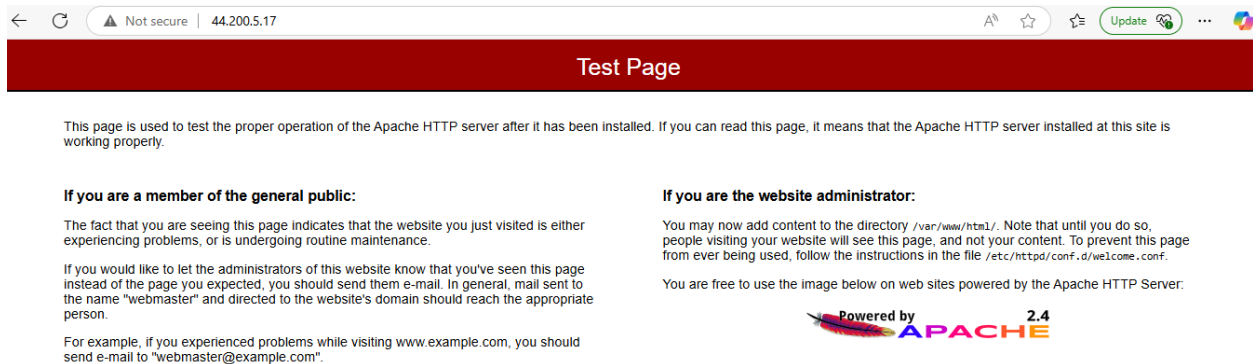
44.200.5.17 | open address

Private IPv4 addresses

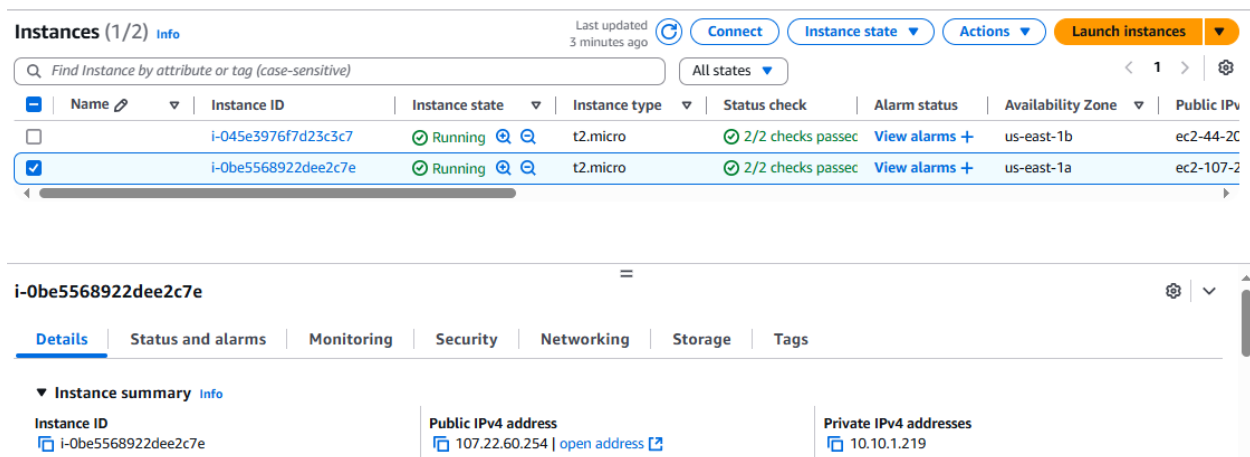
10.10.2.70

Let's make sure we can access the Apache servers!

Copy the public ip 44.200.5.17 of instance1 and past in web browser



Copy the public ip 107.22.60.254 of instance2 and past in web browser



-----Web tier is done-----

Now we can move on to the application tier.

Application Tier

This should be a lot quicker to set up for us given that we're repeating the same process we did for the web tier for the most part but with a few differences.

On the EC2 dashboard, go back to Launch Templates → Create launch template. Let's name the template and add a description. Be sure to check "Provide guidance" on the Auto scaling guidance

section, click on quick start and choose the same free tier OS you used for the web tier- in my case, Amazon Linux.

Let's provide below details

Launch template name: app-tier-template

Template version description: app tier template

Auto Scaling guidance: check the box "Provide guidance to help me set up a template that I can use with EC2 Auto Scaling"

☰ [EC2](#) > [Launch templates](#) > [Create launch template](#)

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - *required*

app-tier-template

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '*', '@'.

Template version description

app tier template

Max 255 chars

Auto Scaling guidance | [Info](#)

Select this if you intend to use this template with EC2 Auto Scaling

☒ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

► **Template tags**

► **Source template**

Launch template contents

On the next Launch template contents section below, choose a free tier OS. I go with Amazon Linux.

Instance type: t2.micro

Key pair (login): multi-tier-project-keypair

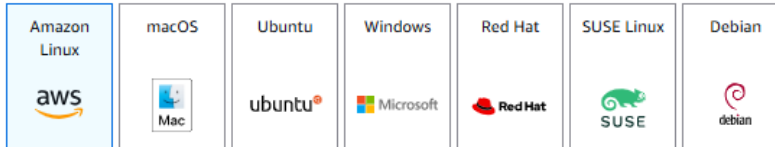
▼ Application and OS Images (Amazon Machine Image) - required [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Search our full catalog including 1000s of application and OS images

Recents

Quick Start



[Browse more AMIs](#)
Including AMIs from
AWS, Marketplace and
the Community

Amazon Machine Image (AMI)

Amazon Linux 2 AMI (HVM) - Kernel 5.10, SSD Volume Type
ami-07a6f770277670015 (64-bit (x86)) / ami-06357f5cb4aea07d8 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

Description

Amazon Linux 2 comes with five years support. It provides Linux kernel 5.10 tuned for optimal performance on Amazon EC2, systemd 219, GCC 7.3, Glibc 2.26, Binutils 2.29.1, and the latest software packages through extras. This AMI is the successor of the Amazon Linux AMI that is now under maintenance only mode and has been removed from this wizard.

Amazon Linux 2 Kernel 5.10 AMI 2.0.20250405.0 x86_64 HVM gp2

Architecture

64-bit (x86) ▼

AMI ID

ami-07a6f770277670015

Publish Date

2025-04-05

Username ⓘ

ec2-user

Verified provider

▼ Instance type [Info](#) | [Get advice](#)

[Advanced](#)

Instance type

t2.micro

Free tier eligible

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Windows base pricing: 0.0162 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour

On-Demand SUSE base pricing: 0.0116 USD per Hour On-Demand RHEL base pricing: 0.026 USD per Hour

On-Demand Linux base pricing: 0.0116 USD per Hour

☐ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

multi-tier-project-keypair ▼

[Create new key pair](#)

For the Network settings section

Firewall (security groups): select existing security group

Security group: app-tier-securitygroup

▼

Network settings

Info

Subnet

Info

Don't include in launch template

▼

↺

Create new subnet

↗

When you specify a subnet, a network interface is automatically added to your template.

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Select existing security group

☐ Create security group

Security groups

Info

Select security groups

▼

app-tier-securitygroup sg-0da0117844fbc3c2

×

VPC: vpc-031c57b95dfed4773

↺

Compare security group rules

►

Advanced network configuration

▼

Storage (volumes)

Info

EB5 Volumes

Hide details

►

Volume 1 (AMI Root) : 8 GiB, EBS, General purpose SSD (gp2)

AMI Volumes are not included in the template unless modified

①

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

×

Add new volume

▼

Resource tags

Info

No resource tags are currently included in this template. Add a resource tag to include it in the launch template.

Add new tag

You can add up to 50 more tags.

►

Advanced details

Info

▼

Summary

Software image (AMI)

Amazon Linux 2 Kernel 5.10 AMI...[read more](#)

ami-07a6f770277670015

Virtual server type (instance type)

t2.micro

Firewall (security group)

app-tier-securitygroup

Storage (volumes)

1 volume(s) - 8 GiB

①

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

×

Cancel

Create launch template

Leave everything else in default and click “Create launch template”.

EC2 > Launch templates > Create launch template

Success

Successfully created `app-tier-template(lt-05b7acce565a2de5)`.

►

Actions log

Next Steps

Launch an instance

With On-Demand Instances, you pay for compute capacity by the second (for Linux, with a minimum of 60 seconds) or by the hour (for all other operating systems) with no long-term commitments or upfront payments. Launch an On-Demand Instance from your launch template.

[Launch instance from this template](#)

Create an Auto Scaling group from your template

Amazon EC2 Auto Scaling helps you maintain application availability and allows you to scale your Amazon EC2 capacity up or down automatically according to conditions you define. You can use Auto Scaling to help ensure that you are running your desired number of Amazon EC2 instances during demand spikes to maintain performance and decrease capacity during lulls to reduce costs.

[Create Auto Scaling group](#)

Create Spot Fleet

A Spot Instance is an unused EC2 instance that is available for less than the On-Demand price. Because Spot Instances enable you to request unused EC2 instances at steep discounts, you can lower your Amazon EC2 costs significantly. The hourly price for a Spot Instance (of each instance type in each Availability Zone) is set by Amazon EC2, and adjusted gradually based on the long-term supply of and demand for Spot Instances. Spot instances are well-suited for data-analysis, batch jobs, background processing, and optional tasks.

[Create Spot Fleet](#)

View launch templates

let's Create app tier Auto Scaling group.

Select the template(app-tier-template) → Actions → create auto scaling group

The screenshot shows the AWS Management Console 'Launch Templates' page. On the left, the navigation pane includes 'EC2', 'Dashboard', 'EC2 Global View', 'Events', 'Instances', 'Instance Types', 'Launch Templates' (highlighted), 'Spot Requests', and 'Savings Plans'. The main content area is titled 'Launch Templates (1/2) Info'. It contains a table with two launch templates: 'app-tier-template' (ID: lt-05b7acce565a2de5) and 'web-tier-template' (ID: lt-0a3db827d5495d1ce). The 'app-tier-template' is selected. An 'Actions' menu is open for the selected template, showing options like 'Launch instance from template', 'Modify template (Create new version)', 'Delete template', 'Delete template version', 'Set default version', 'Manage tags', 'Create Spot Fleet', 'Create Auto Scaling group' (highlighted), and 'View details'. A 'Create launch template' button is visible in the top right corner.

Provide below details Name

Auto Scaling group name: app-tier-auto-scaling-group

Launch template: app-tier-template

Click on Next

The screenshot shows the 'Create Auto Scaling group' wizard in the AWS Management Console, specifically Step 1: 'Choose launch template'. The left sidebar shows the progress of the steps: Step 1 (selected), Step 2, Step 3 (optional), Step 4 (optional), Step 5 (optional), Step 6 (optional), Step 7, and Review. The main content area is titled 'Choose launch template Info'. It includes a 'Name' section with the 'Auto Scaling group name' field set to 'app-tier-auto-scaling-group'. Below this is a 'Launch template' section with a dropdown menu showing 'app-tier-template' and a 'Version' dropdown set to 'Default (1)'. A warning message states: 'For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.' Below the warning, there are links to 'Create a launch template' and 'Create a launch template version'. At the bottom, there is a summary table with the following details:

Create a launch template version		
Description app tier template	Launch template app-tier-template lt-05b7acce565a2de5	Instance type t2.micro
AMI ID ami-07a6f770277670015	Security groups -	Request Spot Instances No
Key pair name multi-tier-project-keypair	Security group IDs sg-0da0117844fbcd3c2	
Additional details		
Storage (volumes) -	Date created Fri Apr 11 2025 15:21:45 GMT+0100 (British Summer Time)	

At the bottom right, there are 'Cancel' and 'Next' buttons.

Select the VPC (multi-tier-vpc) for this project and the two private app tier subnets.
Click "Next"

EC2 > Auto Scaling groups > Create Auto Scaling group Q

Step 1

Choose launch template

Step 2

Choose instance launch options

Step 3 - optional

Integrate with other services

Step 4 - optional

Configure group size and scaling

Step 5 - optional

Add notifications

Step 6 - optional

Add tags

Step 7

Review

Choose instance launch options Info

Choose the VPC network environment that your instances are launched into, and customize the instance types and purchase options.

Instance type requirements Info

Override launch template

You can keep the same instance attributes or instance type from your launch template, or you can choose to override the launch template by specifying different instance attributes or manually adding instance types.

Launch template	Version	Description
app-tier-template ↗ lt-05b7acce565a2de5	Default	app tier template

Instance type
t2.micro

Network Info

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC
Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-031c57b95dfed4773 (multi-tier-vpc)
10.10.0.0/16 [↗](#) [↻](#)

[Create a VPC \[↗\]\(#\)](#)

Availability Zones and subnets

Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets [↻](#)

us-east-1b | subnet-0441fa75497cf7de9 (app-tier-private-2) [✕](#)
10.10.4.0/24

us-east-1a | subnet-03082b07ed85f59e6 (app-tier-private-1) [✕](#)
10.10.3.0/24

[Create a subnet \[↗\]\(#\)](#)

Availability Zone distribution - new

Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

☒ **Balanced best effort**

If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.

☐ **Balanced only**

If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy Availability Zone to preserve balanced distribution.

Cancel

Skip to review

Previous

Next

We're going to go with No load balancer for the app tier and click "Next".

Step 1

Choose launch template

Step 2

Choose instance launch options

Step 3 - optional

Integrate with other services

Step 4 - optional

Configure group size and scaling

Step 5 - optional

Add notifications

Step 6 - optional

Add tags

Step 7

Review

Integrate with other services - *optional* [Info](#)

Use a load balancer to distribute network traffic across multiple servers. Enable service-to-service communications with VPC Lattice. Shift resources away from impaired Availability Zones with zonal shift. You can also customize health check replacements and monitoring.

Load balancing [Info](#)

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

☒ **No load balancer**
Traffic to your Auto Scaling group will not be fronted by a load balancer.

☐ **Attach to an existing load balancer**
Choose from your existing load balancers.

☐ **Attach to a new load balancer**
Quickly create a basic load balancer to attach to your Auto Scaling group.

VPC Lattice integration options [Info](#)

To improve networking capabilities and scalability, integrate your Auto Scaling group with VPC Lattice. VPC Lattice facilitates communications between AWS services and helps you connect and manage your applications across compute services in AWS.

Select VPC Lattice service to attach

☒ **No VPC Lattice service**
VPC Lattice will not manage your Auto Scaling group's network access and connectivity with other services.

☐ **Attach to VPC Lattice service**
Incoming requests associated with specified VPC Lattice target groups will be routed to your Auto Scaling group.

[Create new VPC Lattice service](#)

Application Recovery Controller (ARC) zonal shift - *new* [Info](#)

During an Availability Zone impairment, target instance launches towards other healthy Availability Zones.

☐ **Enable zonal shift**
New instance launches will be retargeted towards healthy Availability Zones until the zonal shift is canceled.

Health checks

Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.

EC2 health checks

[Always enabled](#)

Additional health check types - *optional* [Info](#)

☐ **Turn on Elastic Load Balancing health checks**
Elastic Load Balancing monitors whether instances are available to handle requests. When it reports an unhealthy instance, EC2 Auto Scaling can replace it on its next periodic check.

☐ **Turn on VPC Lattice health checks**
VPC Lattice can monitor whether instances are available to handle requests. If it considers a target as failed a health check, EC2 Auto Scaling replaces it after its next periodic check.

☐ **Turn on Amazon EBS health checks**
EBS monitors whether an instance's root volume or attached volume stalls. When it reports an unhealthy volume, EC2 Auto Scaling can replace the instance on its next periodic health check.

Health check grace period [Info](#)

This time period delays the first health check until your instances finish initializing. It doesn't prevent an instance from terminating when placed into a non-running state.

seconds

Cancel

Skip to review

Previous

Next

The Group size will be the same as the web tier- Desired and Min capacity will be 2 and Max capacity will be 4. Click "Next".

- Step 1
● Choose launch template
- Step 2
● Choose instance launch options
- Step 3 - optional
● Integrate with other services
- Step 4 - optional
● **Configure group size and scaling**
- Step 5 - optional
● Add notifications
- Step 6 - optional
● Add tags
- Step 7
● Review

Configure group size and scaling - optional [Info](#)

Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size [Info](#)

Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances) ▼

Desired capacity

Specify your group size.

2

Scaling [Info](#)

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

2

Equal or less than desired capacity

Max desired capacity

4

Equal or greater than desired capacity

Automatic scaling - optional

Automatic scaling - optional

Choose whether to use a target tracking policy [Info](#)

You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☒ No scaling policies

Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☐ Target tracking scaling policy

Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Instance maintenance policy [Info](#)

Control your Auto Scaling group's availability during instance replacement events. This includes health checks, instance refreshes, maximum instance lifetime features and events that happen automatically to keep your group balanced, called rebalancing events.

Choose a replacement behavior depending on your availability requirements

Mixed behavior

☒ No policy

For rebalancing events, new instances will launch before terminating others. For all other events, instances terminate and launch at the same time.

Prioritize availability

☐ Launch before terminating

Launch new instances and wait for them to be ready before terminating others. This allows you to go above your desired capacity by a given percentage and may temporarily increase costs.

Control costs

☐ Terminate and launch

Terminate and launch instances at the same time. This allows you to go below your desired capacity by a given percentage and may temporarily reduce availability.

Flexible

☐ Custom behavior

Set custom values for the minimum and maximum amount of available capacity. This gives you greater flexibility in setting how far below and over your desired capacity EC2 Auto Scaling goes when replacing instances.

Additional capacity settings

Capacity Reservation preference | [Info](#)

Select whether you want Auto Scaling to launch instances into an existing Capacity Reservation or Capacity Reservation resource group.

☒ **Default**
Auto Scaling uses the Capacity Reservation preference from your launch template.

☐ **None**
Instances will not be launched into a Capacity Reservation.

☐ **Capacity Reservations only**
Instances will only be launched into a Capacity Reservation. If capacity isn't available, the instances fail to launch.

☐ **Capacity Reservations first**
Instances will attempt to launch into a Capacity Reservation first. If capacity isn't available, instances will run in On-Demand capacity.

Additional settings

Instance scale-in protection

If protect from scale in is enabled, newly launched instances will be protected from scale in by default.

☐ Enable instance scale-in protection

Monitoring | [Info](#)

☐ Enable group metrics collection within CloudWatch

Default instance warmup | [Info](#)

The amount of time that CloudWatch metrics for new instances do not contribute to the group's aggregated instance metrics, as their usage data is not reliable yet.

☐ Enable default instance warmup

[Cancel](#)
[Skip to review](#)
[Previous](#)
[Next](#)

Click next few screens until you get to the Review screen and click “Create auto scaling group”.

app-tier-auto-scaling-group created successfully below.

EC2

>

Auto Scaling groups

app-tier-auto-scaling-group created successfully

×

Auto Scaling groups (1/2)

Info

Launch configurations

Launch templates

Actions

Create Auto Scaling group

Search your Auto Scaling groups

< 1 >

⚙

☐	Name	Launch template/configuration	Instances	Status	Desired capacity	Min	Max	Availability Zones
☒	app-tier-auto-scaling-group	app-tier-template Version Default	2	-	2	2	4	us-east-1a, us-east-1b
☐	multi-tier-auto-scaling-group	web-tier-template Version Default	2	-	2	2	4	us-east-1a, us-east-1b

It was successfully created but let’s also check to see of the instances are up and running.

EC2

>

Instances

Instances (4)

Info

Find Instance by attribute or tag (case-sensitive)

All states

Last updated 1 minute ago

Connect

Instance state

Actions

Launch instances

< 1 >

⚙

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...
☐		i-045e3976f7d23c3c7	Running	t2.micro	2/2 checks passec	View alarms +	us-east-1b	ec2-44-200-5-17.comp...	44.200.5.17
☐		i-0d0c33a91ea85e97d	Running	t2.micro	2/2 checks passec	View alarms +	us-east-1b	ec2-44-201-17-227.co...	44.201.17.227
☐		i-0be5568922dee2c7e	Running	t2.micro	2/2 checks passec	View alarms +	us-east-1a	ec2-107-22-60-254.co...	107.22.60.254
☐		i-0bab3a4055015b383	Running	t2.micro	2/2 checks passec	View alarms +	us-east-1a	ec2-54-157-189-158.co...	54.157.189.158

It's a good idea to name the instances for good organization.

Instances (4) Info

Last updated less than a minute ago

Find Instance by attribute or tag (case-sensitive)

All states

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability ...	Public IPv4 DNS	Public IPv...
☐	web-tier-public-2	i-045e3976f7d23c3c7	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1b	ec2-44-200-5-17.comp...	44.200.5.1
☐	app-tier-private-2	i-0d0c33a91ea85e97d	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1b	ec2-44-201-17-227.co...	44.201.17.
☐	web-tier-public-1	i-0be5568922dee2c7e	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	ec2-107-22-60-254.co...	107.22.60.
☐	app-tier-private-1	i-0bab3a4055015b383	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	ec2-54-157-189-158.co...	54.157.18

let's SSH into the web tier's instance 1 to verify we can ping the app tier from the web tier. Select the corresponding instance and click "Connect". Click on EC2 Instance Connect, and click Connect at the bottom.

Once we have access to the instance, run the ping command for the private IP address of the app tier's private instance 1.

ping 10.10.3.135

EC2 > Instances

Instances (1/4) Info

Last updated less than a minute ago

Find Instance by attribute or tag (case-sensitive)

All states

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
☐	web-tier-public-2	i-045e3976f7d23c3c7	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1b	ec2-44-200-5-17.comp...	44.200
☐	app-tier-private-2	i-0d0c33a91ea85e97d	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1b	ec2-44-201-17-227.co...	44.201
☒	web-tier-public-1	i-0be5568922dee2c7e	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	ec2-107-22-60-254.co...	107.22
☐	app-tier-private-1	i-0bab3a4055015b383	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	ec2-54-157-189-158.co...	54.157

EC2 > Instances > i-0be5568922dee2c7e > Connect to instance

Connect to instance Info

Connect to your instance i-0be5568922dee2c7e (web-tier-public-1) using any of these options

EC2 Instance Connect | Session Manager | SSH client | EC2 serial console

Instance ID
i-0be5568922dee2c7e (web-tier-public-1)

Connection Type

☒ Connect using EC2 Instance Connect
Connect using the EC2 Instance Connect browser-based client, with a public IPv4 or IPv6 address.

☐ Connect using EC2 Instance Connect Endpoint
Connect using the EC2 Instance Connect browser-based client, with a private IPv4 address and a VPC endpoint.

☒ Public IPv4 address
107.22.60.254

☐ IPv6 address

Username
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user.

ec2-user

Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel Connect


```

      #_
     /  \  #####
    /    \ #####\
   /      \####|
  /        \#/
 /          V~' '->
/
/m/' '->

Amazon Linux 2
AL2 End of Life is 2026-06-30.
A newer version of Amazon Linux is available!
Amazon Linux 2023, GA and supported until 2028-03-15.
https://aws.amazon.com/linux/amazon-linux-2023/

[ec2-user@ip-10-10-1-219 ~]$ hostname
ip-10-10-1-219.ec2.internal
[ec2-user@ip-10-10-1-219 ~]$
[ec2-user@ip-10-10-1-219 ~]$ ping 10.10.3.135
PING 10.10.3.135 (10.10.3.135) 56(84) bytes of data.
64 bytes from 10.10.3.135: icmp_seq=1 ttl=255 time=0.630 ms
64 bytes from 10.10.3.135: icmp_seq=2 ttl=255 time=1.36 ms
64 bytes from 10.10.3.135: icmp_seq=3 ttl=255 time=0.484 ms
64 bytes from 10.10.3.135: icmp_seq=4 ttl=255 time=0.838 ms
64 bytes from 10.10.3.135: icmp_seq=5 ttl=255 time=0.650 ms
64 bytes from 10.10.3.135: icmp_seq=6 ttl=255 time=0.730 ms
^C
--- 10.10.3.135 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5090ms
rtt min/avg/max/mdev = 0.484/0.782/1.365/0.283 ms
[ec2-user@ip-10-10-1-219 ~]$
[ec2-user@ip-10-10-1-219 ~]$

```

i-0be5568922dee2c7e (web-tier-public-1)

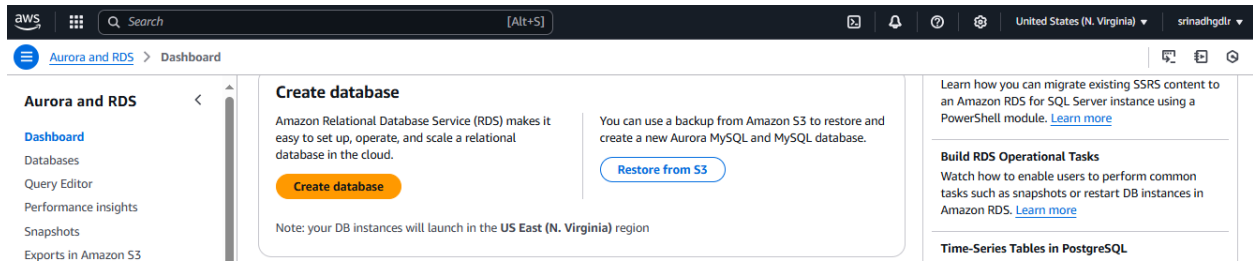
PublicIPs: 107.22.60.254 PrivateIPs: 10.10.1.219

Type Ctrl C to stop the pinging in the server
It is working now.

Now we can start the last tier – database tier

Database Tier

Search for RDS on the AWS console and click “Create database”.



Select Standard create and MySQL

[Aurora and RDS](#) > Create database

Create database [Info](#)

Choose a database creation method

☒ **Standard create**
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ **Easy create**
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Engine options

Engine type [Info](#)

☐ Aurora (MySQL Compatible)


☐ Aurora (PostgreSQL Compatible)


☒ **MySQL**


☐ PostgreSQL


☐ MariaDB

☐ Oracle

☐ Microsoft SQL Server


☐ IBM Db2
IBM Db2

Edition

☒ **MySQL Community**

Engine version [Info](#)

View the engine versions that support the following database features.

▼ **Hide filters**

☐ Show only versions that support the Multi-AZ DB cluster [Info](#)
Create a Multi-AZ DB cluster with one primary DB instance and two readable standby DB instances. Multi-AZ DB clusters provide up to 2x faster transaction commit latency and automatic failover in typically under 35 seconds.

☐ Show only versions that support the Amazon RDS Optimized Writes [Info](#)
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

Engine version

MySQL 8.0.40 ▼

☐ **Enable RDS Extended Support** [Info](#)
Amazon RDS Extended Support is a [paid offering](#). By selecting this option, you consent to being charged for this offering if you are running your database major version past the RDS end of standard support date for that version. Check the end of standard support date for your major version in the [RDS for MySQL documentation](#).

Select free tier for the templates and name the database as database-1
You can keep admin for the master username but be prepared to make a master password.

Templates

Choose a sample template to meet your use case.

☐ Production
Use defaults for high availability and fast, consistent performance.

☐ Dev/Test
This instance is intended for development use outside of a production environment.

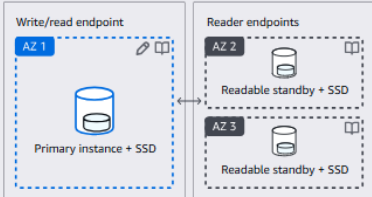
☒ Free tier
Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS. [Info](#)

Availability and durability

Deployment options [Info](#)
Choose the deployment option that provides the availability and durability needed for your use case. AWS is committed to a certain level of uptime depending on the deployment option you choose. Learn more in the [Amazon RDS service level agreement \(SLA\)](#).

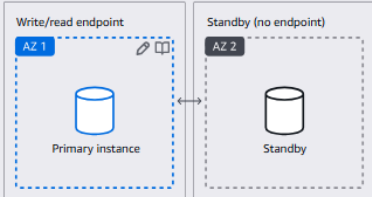
☐ Multi-AZ DB cluster deployment (3 instances)
Creates a primary DB instance with two readable standbys in separate Availability Zones. This setup provides:

- 99.95% uptime
- Redundancy across Availability Zones
- Increased read capacity
- Reduced write latency



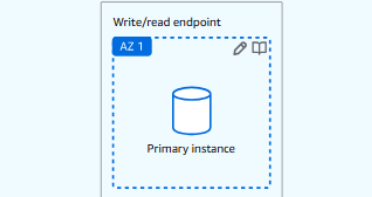
☐ Multi-AZ DB instance deployment (2 instances)
Creates a primary DB instance with a non-readable standby instance in a separate Availability Zone. This setup provides:

- 99.95% uptime
- Redundancy across Availability Zones



☒ Single-AZ DB instance deployment (1 instance)
Creates a single DB instance without standby instances. This setup provides:

- 99.5% uptime
- No data redundancy



[Aurora and RDS](#) > Create database

Settings

DB instance identifier [Info](#)
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

database-1

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

▼ **Credentials Settings**

Master username [Info](#)
Type a login ID for the master user of your DB instance.

admin

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - *most secure*
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed
Create your own password or have RDS create a password that you manage.

☐ Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

Password strength Weak

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' " @

Confirm master password [Info](#)

For Instance configuration let's make sure burstable classes is checked and then skip to the connectivity section to select the VPC for the project.

Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

▼ Hide filters

- ☒ Show instance classes that support Amazon RDS Optimized Writes [Info](#)
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.
- ☐ Include previous generation classes
- ☐ Standard classes (includes m classes)
- ☐ Memory optimized classes (includes r and x classes)
- ☒ Burstable classes (includes t classes)

db.t4g.micro
2 vCPUs 1 GiB RAM Network: Up to 2,085 Mbps

Click “Create new” for VPC security group and name it as database-tier-security-group. Leave everything else at default and scroll down and click “Create database” at the bottom.

Connectivity [Info](#)

Compute resource

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

- ☒ Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.
- ☐ Connect to an EC2 compute resource
Set up a connection to an EC2 compute resource for this database.

Network type [Info](#)

To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

- ☒ IPv4
Your resources can communicate only over the IPv4 addressing protocol.
- ☐ Dual-stack mode
Your resources can communicate over IPv4, IPv6, or both.

Virtual private cloud (VPC) [Info](#)

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

multi-tier-vpc (vpc-031c57b95dfed4773)
6 Subnets, 2 Availability Zones

Only VPCs with a corresponding DB subnet group are listed.

After a database is created, you can't change its VPC.

DB subnet group [Info](#)

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

Create new DB Subnet Group

Public access [Info](#)

- ☐ Yes
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.
- ☒ No

VPC security group (firewall) [Info](#)

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

- ☐ Choose existing
Choose existing VPC security groups
- ☒ Create new
Create new VPC security group

New VPC security group name

database-tier-securitygroup

Availability Zone [Info](#)

No preference

RDS Proxy

RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.

- ☐ Create an RDS Proxy [Info](#)
RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see [Amazon RDS Proxy pricing](#).

Certificate authority - optional [Info](#)

Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

rds-ca-rsa2048-g1 (default)
Expiry: May 26, 2061

If you don't select a certificate authority, RDS chooses one for you.

► Additional configuration

Estimated monthly costs

The Amazon RDS Free Tier is available to you for 12 months. Each calendar month, the free tier will allow you to use the Amazon RDS resources listed below for free:

- 750 hrs of Amazon RDS in a Single-AZ db.t2.micro, db.t3.micro or db.t4g.micro Instance.
- 20 GB of General Purpose Storage (SSD).
- 20 GB for automated backup storage and any user-initiated DB Snapshots.

[Learn more about AWS Free Tier.](#)

When your free usage expires or if your application use exceeds the free usage tiers, you simply pay standard, pay-as-you-go service rates as described in the [Amazon RDS Pricing page.](#)

You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that you use with AWS services.

[Cancel](#)

[Create database](#)

VPC dashboard < **Security Groups (1/5)** [Info](#) [Actions](#) [Export security groups to CSV](#) [Create security group](#)

Q Find resources by attribute or tag

Name	Security group ID	Security group name	VPC ID	Description
-	sg-04fe31b763f20b828	default	vpw-031c57b95dfe4773	default VPC security group
-	sg-0b4024219fb57905	default	vpw-0f982f9178eb91ad4	default VPC security group
-	sg-0d6707673c4d99fc1	database-tier-securitygroup	vpw-031c57b95dfe4773	Created by RDS management console
-	sg-0da0117844fbcd3c2	app-tier-securitygroup	vpw-031c57b95dfe4773	Allow SSH access
-	sg-0ae1b5fa1429f71e3	multi-tier-project-securitygroup	vpw-031c57b95dfe4773	Allow SSH access

sg-0d6707673c4d99fc1 - database-tier-securitygroup

[Details](#) [Inbound rules](#) [Outbound rules](#) [Sharing - new](#) [VPC associations - new](#) [Tags](#)

Inbound rules (1) [Manage tags](#) [Edit inbound rules](#)

Q Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sgr-02f61efb963cdcc05	IPv4	MySQL/Aurora	TCP	3306	77.99.97.209/32	-

Go the VPC dashboard in the AWS console and click on security groups. Select the security group for the database tier, click “Actions”, and select “Edit inbound rules”.

Delete the security group rule, click “Add rule”, select MySQL/Aurora for the Type, and select the app-tier security group for the Source. Click “Save rules”.

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules [Info](#)

Security group rule ID

Type [Info](#)

Protocol [Info](#)

Port range [Info](#)

Source [Info](#)

Description - optional [Info](#)

-

MySQL/Aurora

TCP

3306

Custom

Q sg-Od6707673c4d99fc1

X

Use: "sg-Od6707673c4d99fc1"

CIDR blocks

Security Groups

database-tier-securitygroup | sg-Od6707673c4d99fc1

Prefix lists

Add rule

Preview changes

Delete

Save rules

Actions

Export security groups to CSV

Create security group

Security Groups (1/5) [Info](#)

Find resources by attribute or tag

	Name	Security group ID	Security group name	VPC ID	Description	
☒	-	sg-Od6707673c4d99fc1	database-tier-securitygroup	vpc-031c57b95dfed4773	Created by RDS management console	2
☐	-	sg-04fe31b763f20b828	default	vpc-031c57b95dfed4773	default VPC security group	2
☐	-	sg-0b4024219fbf57905	default	vpc-0f982f9178eb91a4f	default VPC security group	2
☐	-	sg-0da0117844fbcd3c2	app-tier-securitygroup	vpc-031c57b95dfed4773	Allow SSH access	2

sg-Od6707673c4d99fc1 - database-tier-securitygroup

Details **Inbound rules** Outbound rules Sharing - new VPC associations - new Tags

Inbound rules (1)

Q Search

< 1 >

⚙

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
☐	-	sgr-04d699165f76bfe9c	-	MySQL/Aurora	TCP	3306	sg-Od6707673c4d99fc1 / database-tier-securitygroup	-

VPC > Security Groups > sg-Od6707673c4d99fc1 - database-tier-securitygroup > Edit inbound rules

🔍 ⌂ 🖨

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules [Info](#)

Security group rule ID

Type [Info](#)

Protocol [Info](#)

Port range [Info](#)

Source [Info](#)

Description - optional [Info](#)

sg-O4d699165f76bfe9c

MySQL/Aurora

TCP

3306

Custom

Q sg-0da0117844fbcd3c2

X

Use: "sg-0da0117844fbcd3c2"

CIDR blocks

Security Groups

app-tier-securitygroup | sg-0da0117844fbcd3c2

Prefix lists

Add rule

Preview changes

Delete

Save rules

Security Groups (1/5) Info

Find resources by attribute or tag						
☒	Name	Security group ID	Security group name	VPC ID	Description	
☒	-	sg-0d6707673c4d99fc1	database-tier-securitygroup	vpc-031c57b95dfed4773	Created by RDS management console	2
☐	-	sg-04fe31b763f20b828	default	vpc-031c57b95dfed4773	default VPC security group	2
☐	-	sg-0b40242191bf57905	default	vpc-0f982f9178eb91a4f	default VPC security group	2
☐	-	sg-0da0117844fbcd3c2	app-tier-securitygroup	vpc-031c57b95dfed4773	Allow SSH access	2

sg-0d6707673c4d99fc1 - database-tier-securitygroup

Details

Inbound rules

Outbound rules

Sharing - new

VPC associations - new

Tags

Inbound rules (1)

Q Search

Name

Security group rule ID

IP version

Type

Protocol

Port ra...

Source

Description

-

sgr-04d699165f76bfe9c

-

MySQL/Aurora

TCP

3306

sg-0da0117844fbcd3c2 / app-tier-securitygroup

-

Manage tags

Edit inbound rules

<1>

Successfully created database now

Aurora and RDS > Databases

Successfully created database database-1

You can use settings from database-1 to simplify configuration of suggested database add-ons while we finish creating your DB for you.

View connection details

Databases (1)

Filter by databases									
DB identifier	Status	Role	Engine	Region ...	Size	Recommendations	CPU	Current activi	
database-1	Available	Instance	MySQL Co...	us-east-1a	db.t4g.micro	-			

database-1

Summary

DB identifier
database-1

CPU
4.71%

Status
Available

Class
db.t4g.micro

Role
Instance

Current activity
0 Connections

Engine
MySQL Community

Region & AZ
us-east-1a

Recommendations

Connectivity & security | Monitoring | Logs & events | Configuration | Zero-ETL integrations | Maintenance & backups | Data migrations - new | Tags | Recommendations

Instance			
Configuration			
DB instance ID database-1	Instance class db.t4g.micro	Encryption Enabled	Monitoring type Database Insights - Standard
Engine version 8.0.40	vCPU 2	AWS KMS key aws/rds	Performance Insights Disabled
RDS Extended Support Disabled	RAM 1 GB	Storage type General Purpose SSD (gp2)	Enhanced Monitoring Disabled
DB name -	Availability	Storage 20 GiB	DevOps Guru Disabled
License model General Public License	Master username admin	Provisioned IOPS -	
Option groups default:mysql-8-0 In sync	Master password *****	Storage throughput -	
Amazon Resource Name (ARN) arn:aws:rds:us-east-1:284514300790:db:database-1	IAM DB authentication Not enabled	Storage autoscaling Enabled	
Resource ID db-TIW7SRUNRMOXGFAQJHNYRV2P5Q	Multi-AZ No	Maximum storage threshold 1000 GiB	
Created time April 11, 2025, 16:31 (UTC+01:00)	Secondary Zone -	Storage file system configuration Current	
DB instance parameter group default.mysql8.0 In sync			
Deletion protection Disabled			